

Two-Timescale Potency: Splitting Potency into Instantaneous and Integrated Components

Design discussion notes

2026-09-02

Scope

This note develops a model in which **potency** (Nature 2024, Fig. 3d) is decomposed into an *instantaneous* component — how much a TF bound **right now, on this molecule** promotes promoter activity — and an *integrated* component — how much the molecule’s **time-averaged** binding history promotes activity. A single mixing parameter $w \in [0, 1]$ sets the split.

The motivation is a specific objection to an earlier proposal (in 260717_ising_intuition.md §7) that potency could be read off as an equilibrium coupling J_{prom} between promoter state and same-molecule TF occupancy. **That proposal is wrong for this system, for two independent reasons**, both visible in Fig. 3:

1. The functional form is wrong (Fig. 3d): an equilibrium energy coupling predicts a *convex* occupancy–activity curve; the data are straight-to-concave (§2.1).
2. It has no representation of history (Fig. 3h): promoter activity depends on the number of *available* sites and not only on the number *currently bound* (§2.2).

The model below fixes both, reproduces the published potency exactly as a limiting case, and makes w measurable from data already in hand.

1. Notation

symbol	meaning
K	TetO copy number of the construct (0–8, or 0–9 for opJS5)
n	TF occupancy on this molecule, in this snapshot (v5’s <code>n_tf_teto</code>)
$\langle n \rangle$	mean occupancy for this molecule’s <i>condition</i> (K, dox, AD) = Fig. 3c/3i
P	probability this molecule’s promoter is active (v5 region-anchored promoter call)
$\bar{P}$	condition-level mean of P = “fraction of promoters active”, Fig. 3b/3d
p_0	basal activity at zero occupancy (intercept)

symbol	meaning
A	odds of being active, $A = P/(1 - P)$; used only for the hyperbolic-link alternative
k_{pot}	potency (units: drive per bound TF)
w	integrated fraction of potency (0 = purely instantaneous, 1 = purely historical)
τ_{int}	promoter integration timescale
τ_c	correlation (residence) timescale of TF occupancy on one molecule

2. What Fig. 3 constrains

2.1 Panel 3d fixes the *link function*: line-like, definitely not energy-additive

Panel 3d plots $\bar{P}$ against $\langle n \rangle$: $\bar{P}$ runs from ≈ 0.03 at $\langle n \rangle = 0$ to ≈ 0.47 at $\langle n \rangle \approx 5$.

Consider three candidate couplings.

Energy-additive (Ising / Boltzmann). Each bound TF adds a fixed free energy stabilizing the active state, so the *log-odds* are linear:

$$\text{logit } \bar{P} = h_{\text{prom}} + J \langle n \rangle.$$

Fit to the endpoints: $\text{logit}(0.03) = -3.48$, $\text{logit}(0.47) = -0.12$, so $J \approx 3.36/5 \approx 0.67$. Evaluate at the midpoint $\langle n \rangle = 2.5$: $\text{logit } \bar{P} = -1.80 \Rightarrow \bar{P} \approx 0.14$. A straight line between the endpoints would give 0.25. **So this model predicts pronounced upward (convex) curvature — the midpoint should sit far below the chord.**

Rate-additive (the paper’s Fig. 3f additive activation model). Each bound TF independently adds to the promoter’s opening *rate*, $k_{\text{on}} \propto \text{TF}_{\text{occ}}$, giving

$$\bar{P} = \frac{1}{1 + \frac{k_{\text{off}}}{k_{\text{on}} \cdot \text{TF}_{\text{occ}}}} = \frac{A}{1 + A}, \quad A = k_0 + k_{\text{pot}} \langle n \rangle,$$

i.e. the *odds* (not the log-odds) are linear in occupancy. This is hyperbolic in $\langle n \rangle$: mildly **concave**, essentially straight while $A \lesssim 1$, and it is what `compute_potency.py`’s `saturation_model_mechanistic` already implements.

Identity (a straight line). $\bar{P} = p_0 + k_{\text{pot}} \langle n \rangle$, with no saturation over the observed range.

Ranking these against the data:

link	$\bar{P}$ vs $\langle n \rangle$	verdict
energy-additive (logistic)	strongly convex	rejected — predicts ≈ 0.14 at mid-occupancy where the data sit near 0.25–0.3

link	$\bar{P}$ vs $\langle n \rangle$	verdict
rate-additive (hyperbolic)	mildly concave	fits ($r^2 = 0.96$), but <i>worse than a straight line</i>
identity (linear)	straight	best empirical fit ; adopted here

Two conclusions, at different confidence levels.

Robust: the curve is *not convex*, which rejects the energy-additive/Boltzmann coupling regardless of how the concave-vs-straight question resolves. This is the conclusion that kills the earlier J_{prom} proposal, and it does not depend on distinguishing the other two links.

Empirical: a straight line fits better than the hyperbola, so **this note uses the identity link** and treats eventual saturation as outside the measured range. Nothing below depends on reproducing the hyperbola exactly; the requirement is only that the relation be *line-like*, which it is. Conveniently, the identity link also makes the central result of §4 *exact* rather than approximate.

The linearity itself remains the open mechanistic puzzle, exactly as in the original analysis. A probability cannot be linear in a driving variable forever, so linearity means the system sits far from saturation across the entire accessible range — which is a statement worth explaining, not an explanation. Diagnostic for future ADs: a *stronger* AD pushes $\bar{P}$ higher and should therefore be the first place linearity breaks. **If a new AD reaches high $\bar{P}$ and is still straight, that is a real result**; if it bends, the bend’s shape identifies the link function that the current ADs are too weak to reveal. Either outcome is informative, so this is worth checking per-AD rather than pooling.

2.2 Panel 3h requires *two* occupancy terms

Panel 3h stratifies “fraction of promoters active” by the number of TFs bound (0–5) as a function of TetO number. Two effects are simultaneously present:

- At fixed K , activity rises with n (the curves are ordered by shade).
- **At fixed n , activity rises with K** (each curve slopes upward). A molecule with 4 of 8 sites bound is more likely active than one with 4 of 4 bound.

The second effect is the one that no single-snapshot equilibrium model can produce. In equilibrium, conditioning on the configuration is sufficient: $P(\text{active} \mid \sigma_{\text{TF}})$ is a function of σ_{TF} alone, so it cannot depend on how many *other* sites merely exist.

Two further facts sharpen this into strong evidence, both established from the construct design (all opJS4 variants are exactly 630 bp; operator coordinates are shared and absolute — 4x is the *first four* positions of 8x; the promoter sits at low coordinates so the proximal operator is at 282 in every construct; spacer blocks are CG-free scramble):

- **A cell-to-cell “permissiveness” confound predicts the opposite sign.** If some cells are globally more permissive (more binding *and* more activity), then in a 4x construct “4 bound” is the top of the occupancy distribution and so is enriched for permissive cells, whereas in 8x “4 bound” is typical. That predicts $P(\text{active} \mid 4/4) > P(\text{active} \mid 4/8)$ — the reverse of the observation.

- **A distance-decay coupling also predicts the opposite sign.** In 4x, four bound sites *must* be the four promoter-proximal ones; in 8x they are on average more distal. If proximal binding is more effective, again $4/4 > 4/8$.

Both of the obvious alternatives work *against* the observed effect, so controlling for them should make it larger, not smaller. The one equilibrium mechanism with the right sign — a distal gap acting as a sink for nucleosomes that would otherwise cover the promoter — is excluded by the equal-length design: conditioned on *which* sites are bound, a 4x and an 8x molecule present the same physical DNA (the sole loophole being any intrinsic difference in nucleosome affinity between TetO and scramble sequence).

2.3 Panels 3i–3k constrain how potency may depend on condition

Changing dox changes occupancy (3i), but all dox levels **collapse onto one occupancy–activity curve** (3j), and potency is constant across dox while effective TF concentration varies about two-fold (3k).

So the map (occupancy $\rightarrow$ activity) is a function of occupancy alone: it must not acquire a separate explicit dependence on dox or on K . §4 shows the model satisfies this automatically — which is why panel 3j, despite being the panel that *defines* potency, carries **no information about w at all**.

3. The model

3.1 The drive, and why the mixture form is derived rather than assumed

Let $x(t)$ be the (stochastic) TF occupancy on one molecule over time: stationary, mean $\langle n \rangle$, autocorrelation e^{-s/τ_c} . Suppose the promoter responds not to the instantaneous occupancy but to an exponentially filtered version of it with integration time τ_{int} :

$$D(t) = \frac{1}{\tau_{\text{int}}} \int_0^\infty e^{-s/\tau_{\text{int}}} x(t-s) ds.$$

SMF gives us a *snapshot*: we observe $x(t) = n$, not D . The relevant quantity is therefore D 's conditional expectation given what we measured:

$$\mathbb{E}[D \mid x(t) = n] = \langle n \rangle + (n - \langle n \rangle) \frac{1}{\tau_{\text{int}}} \int_0^\infty e^{-s/\tau_{\text{int}}} e^{-s/\tau_c} ds = \frac{\tau_c}{\tau_{\text{int}} + \tau_c} n + \frac{\tau_{\text{int}}}{\tau_{\text{int}} + \tau_c} \langle n \rangle.$$

Defining

$$\boxed{w = \frac{\tau_{\text{int}}}{\tau_{\text{int}} + \tau_c}, \quad \text{equivalently} \quad \frac{w}{1-w} = \frac{\tau_{\text{int}}}{\tau_c}}$$

gives exactly the mixture

$$\boxed{I = (1-w)n + w\langle n \rangle}$$

So the two-timescale form is **not an ad hoc interpolation**: it is the snapshot-conditional expectation of a time-integrated drive, and w is a direct readout of the ratio of the promoter's integration time to the TF's residence time. $w \rightarrow 0$ means the promoter reads instantaneous occupancy; $w \rightarrow 1$ means it reads only the long-run average, which is a property of the *condition*, not of the molecule.

(Approximation used: $\mathbb{E}[f(D) \mid n] \approx f(\mathbb{E}[D \mid n])$, i.e. linear response of the promoter to the drive. This is exactly the rate-additive assumption of §2.1, so the two approximations are the same one.)

3.2 The full model

Combining §2.1’s link with §3.1’s drive:

$$P(\text{active} \mid n, \langle n \rangle) = p_0 + k_{\text{pot}} I = p_0 + k_{\text{pot}} [(1 - w) n + w \langle n \rangle]$$

(If a future AD turns out to saturate (§2.1), swap the link for $P = A/(1 + A)$ with $A = p'_0 + k_{\text{pot}} I$. Everything structural below is unchanged; only §4’s cancellation weakens from exact to first-order and §7’s estimator changes from least squares to a bounded nonlinear fit.)

Four parameters, all ≥ 0 :

parameter	meaning	identified by
p_0	basal activity at zero occupancy	intercept, Fig. 3d at $\langle n \rangle = 0$
k_{pot}	potency — total drive added per unit occupancy	the bulk slope (Fig. 3d)
w	integrated fraction of potency	<i>within-</i> vs <i>between</i> -condition contrast (Fig. 3h)
$\langle n \rangle$	not fitted — supplied per condition	measured (Fig. 3c/3i) or predicted by the array model

Equivalently, in the linearly-parametrized coordinates $k_{\text{fast}} = k_{\text{pot}}(1 - w)$ and $k_{\text{slow}} = k_{\text{pot}}w$:

$$P = p_0 + k_{\text{fast}} n + k_{\text{slow}} \langle n \rangle, \quad k_{\text{pot}} = k_{\text{fast}} + k_{\text{slow}}, \quad w = \frac{k_{\text{slow}}}{k_{\text{fast}} + k_{\text{slow}}}.$$

Fit in $(p_0, k_{\text{fast}}, k_{\text{slow}})$ — linear in the parameters, so the fit is well-behaved and k_{pot}, w are read off afterwards. (Fitting w and k_{pot} directly introduces a product of unknowns for no benefit.)

Special cases.

- $w = 0$: the pure equilibrium/instantaneous model. Promoter state depends only on this molecule’s own current binding. This is what the earlier J_{prom} proposal assumed, and Fig. 3h rejects it.
- $w = 1$: pure history. A depends only on the condition, so **all** molecules in a condition have the same activation probability regardless of their own occupancy — potency is fully real and fully invisible to any same-molecule analysis. This is the limit the objection correctly identified.
- $k_{\text{slow}} = 0$ and logistic link: recovers the original J_{prom} proposal, now doubly falsified.

4. Why w is invisible in bulk: the exact cancellation

This section formalizes the central intuition: *if the promoter integrates over a long time, potency is real but does not appear in any same-molecule coupling.*

Average over all molecules in one condition. Since $\mathbb{E}[n] = \langle n \rangle$ by definition, and P is **linear** in n :

$$\bar{P} = \mathbb{E}[P] = p_0 + k_{\text{pot}} [(1 - w) \mathbb{E}[n] + w \langle n \rangle] = p_0 + k_{\text{pot}} [(1 - w) \langle n \rangle + w \langle n \rangle] = p_0 + k_{\text{pot}} \langle n \rangle.$$

The w terms cancel identically, and with the identity link this is exact — no approximation. The bulk occupancy–activity curve depends only on p_0 and k_{pot} , for *any* value of w .

The intuition in one sentence: a molecule’s own n is a noisy sample of its condition’s $\langle n \rangle$, and averaging that sample over molecules returns $\langle n \rangle$ again — so bulk data cannot tell whether the promoter read the sample or read the mean. Panel h can, because conditioning on n deliberately examines molecules that *differ* from their condition mean, which is exactly the information averaging destroys.

An equivalent way to see it, which also gives the shape of the answer:

$$\underbrace{k_{\text{pot}}}_{\text{Fig. 3d slope}} = \underbrace{k_{\text{pot}} w}_{\text{panel h: slope of each curve in } \langle n \rangle \text{ at fixed } n} + \underbrace{k_{\text{pot}}(1 - w)}_{\text{panel h: spacing between consecutive } n \text{ curves}}$$

Potency is the conserved quantity; w only decides how it is split between the two axes of panel h. Fig. 3d measures the sum and is therefore blind to w ; panel h resolves the decomposition.

Three consequences:

1. **The published potency is k_{pot}** , cleanly, with no w correction. The Fig. 3d slope and `compute_potency.py`’s k_{tf} estimate k_{pot} , and they remain valid regardless of how the fast/slow split turns out. Nothing in the published analysis needs revision.
2. **Panel 3j cannot measure w** . Its collapse across dox is a genuine and important check that potency is a property of the AD/promoter rather than of how occupancy was achieved (§2.3) — and the model satisfies it automatically — but it is silent on w .
3. **Any purely same-molecule coupling estimates $k_{\text{fast}} = k_{\text{pot}}(1 - w)$, not potency.** It *underestimates* potency by exactly the factor $(1 - w)$, and reports zero when $w = 1$. Hence: a same-molecule coupling is not potency, and potency is not a same-molecule coupling. They are two different, both-meaningful projections of the same k_{pot} .

If the link is ever found to be nonlinear (§2.1, e.g. a strong new AD that saturates), the cancellation degrades from exact to first-order: $\mathbb{E}[f(I)] \neq f(\mathbb{E}[I])$ by Jensen, and larger w means less molecule-to-molecule variance in I and hence a smaller Jensen correction. So w would leave a faint bulk signature — in the *curvature* of Fig. 3d, never its slope. Far too weak to estimate w from, but it means a joint bulk + single-molecule fit should then average over the molecule-level distribution of n rather than substituting $\langle n \rangle$.

4.1 Numerical demonstration

`260902_w_regimes_cartoon.py` simulates molecules under $w = 0, 0.25, 1$ (20,000 molecules per copy number, occupancy drawn $\text{Binomial}(K, \langle n \rangle_K / K)$ so the condition mean matches by construction) and confirms both halves of the claim:

w true	bulk intercept	bulk slope	$\hat{w}$ recovered	$\hat{k}_{\text{pot}}$
0.00	0.031	0.090	0.004	0.090
0.25	0.030	0.090	0.254	0.091
1.00	0.030	0.091	1.006	0.091

The bulk line is indistinguishable across regimes (true $k_{\text{pot}} = 0.09$), while w is recovered accurately from the molecule-level regression. The script also emits a figure contrasting the three regimes in bulk (identical) and in panel-h coordinates (qualitatively different).

5. Where w is visible: three tests

All three condition on n and vary $\langle n \rangle$, which is the only way to break the §4 cancellation.

5.1 Copy number at fixed n (Fig. 3h — already measured)

Rearranged, the model predicts that at fixed n ,

$$P(n, K) = \underbrace{[p_0 + k_{\text{pot}}(1 - w)n]}_{\text{intercept in } \langle n \rangle} + k_{\text{pot}}w \cdot \langle n \rangle_K,$$

i.e. **plotting P against $\langle n \rangle_K$ at fixed n should give a family of parallel straight lines**, one per n , with common slope $k_{\text{pot}}w$ and spacing $k_{\text{pot}}(1 - w)$ between consecutive n . That single plot yields both parameters.

Correction (2026-09-04): the “sum rule” is NOT a test. An earlier version of this note and of §6 treated $k_{\text{pot}}w + k_{\text{pot}}(1 - w) = k_{\text{pot}}^{\text{bulk}}$ as a falsifiable check. When both sides are obtained from **one joint fit to the same data it is an algebraic identity**, verified to hold to machine precision on random data. The reason is the §4 cancellation reappearing in the estimator: within an amplicon the N -weighted mean of n is $\langle n \rangle$, so writing $n = \langle n \rangle + d$ makes d within-amplicon centred and therefore N -orthogonal to every amplicon-level function. The design splits into orthogonal between/within blocks, k_{pot} is estimated purely from between-amplicon variation (i.e. it *is* the bulk fit) and k_{fast} purely from within. They cannot disagree. (§6’s version compares two *independent hand readings* off two published panels, which is informative about the figures’ mutual consistency — but it is not a test of the model, and it is not what any joint fit computes.)

The real structural test is PARALLELISM: add an interaction $n \times \langle n \rangle$ and check its coefficient is zero. Non-zero means the lines fan out and the two-timescale form is wrong. `fit_two_timescale_potency.py` reports `k_interact` and its z -score for exactly this. Panel 3h contains this information already — it plots P against K , and the model wants $\langle n \rangle_K$ on x , which is just re-plotting through Fig. 3c. Panel 3h lacks a statistical test; this gives it one.

The three regimes are qualitatively distinguishable by eye in this plot (simulated in `260902_w_regimes_cartoon.py`):

regime	panel h appearance	interpretation
$w = 0$	curves flat in $\langle n \rangle$, maximally separated by n	promoter reads instantaneous occupancy only
$0 < w < 1$	parallel sloping lines, partially separated	mixture; slope/spacing ratio gives $w/(1 - w) = \tau_{\text{int}}/\tau_c$
$w = 1$	curves collapse onto a single sloping line	promoter reads history only; potency fully real, single-molecule signal zero

Note that $w = 1$ is the regime in which potency exists and is measurable in bulk while *no* same-molecule analysis can detect any coupling whatsoever — the case that motivated this note.

5.2 Dox at fixed K and fixed n — the clean test, not yet run

This is the sharpest available experiment and the data already exist. Hold copy number fixed, so the construct is *physically identical*; vary dox, which changes $\langle n \rangle$ (Fig. 3i) with no change whatsoever in geometry, site positions, or sequence. Then compare molecules with the *same* n across dox levels:

$$\text{if } w > 0 : \quad P(\text{active} \mid n, \text{ high dox}) > P(\text{active} \mid n, \text{ low dox})$$

$$\text{if } w = 0 : \quad \text{these are equal.}$$

Why this beats §5.1: the copy-number comparison must contend with *which* sites are bound differing between constructs (§2.2), even though that confound has the wrong sign. The dox comparison has no structural confound at all — same molecule design, same sites, only the occupancy statistics differ. A monotone increase in $P(\text{active} \mid n)$ with dox at fixed K and fixed n is essentially unimpeachable evidence for $w > 0$.

5.3 Fixed exact configuration (v5 per-site calls)

Strengthen §5.1 by conditioning on *which* sites are bound rather than only how many: compare “operators 282/322/362/402 bound” in an 8x construct against the same four sites bound in a 4x construct. Given the equal-length shared-coordinate design (§2.2), those molecules are physically near-identical, so a residual difference cannot be equilibrium physics of a single snapshot. v5 emits per-site occupancy (`site_{name}` columns), so this is a re-tabulation, not new machinery. The old count-based analysis could not run this test.

6. Estimating w from the published figure (illustrative only)

Approximate values read by eye off Fig. 3h/3c, purely to show the arithmetic and check that the model is identifiable and gives a plausible answer. **These are not measurements** — they should be recomputed from the molecule tables.

Take $\langle n \rangle \approx 0.15$ at $K = 2$ and ≈ 4.1 at $K = 9$, so $\Delta \langle n \rangle \approx 4.0$.

Between-condition contrast (the slope), at fixed $n = 0$: $P \approx 0.01 \rightarrow 0.10$ from $K = 2$ to $K = 9$, so $\Delta P \approx 0.09$ and

$$k_{\text{pot}} w \approx 0.09/4.0 \approx 0.023.$$

Within-condition contrast (the spacing), at fixed $K = 9$: $P \approx 0.10 \rightarrow 0.47$ from $n = 0$ to $n = 5$, so $\Delta P \approx 0.37$ and

$$k_{\text{pot}}(1 - w) \approx 0.37/5 \approx 0.074.$$

$$k_{\text{pot}} \approx 0.097, \quad w \approx \frac{0.023}{0.097} \approx 0.23, \quad \frac{\tau_{\text{int}}}{\tau_c} = \frac{w}{1 - w} \approx 0.31.$$

So roughly a quarter of potency would be integrated and three quarters instantaneous, with the promoter integrating over about a third of a TF residence time.

A genuine consistency check falls out of this. k_{pot} estimated from panel h (≈ 0.097) is an *independent* route to the same quantity as the Fig. 3d slope ($\approx (0.47 - 0.03)/4.9 \approx 0.090$). Those agree to about 7%, which is non-trivial support for the model: nothing forced two different panels, analysed two different ways, to give the same potency. Confirming this properly on the real molecule tables is worth doing early — if the two disagree badly, the decomposition is wrong and no amount of fitting will fix it.

w is link-dependent, so §2.1 comes first. The identical arithmetic gives $w \approx 0.14$ under the hyperbolic link and $w \approx 0.59$ under the logistic — a qualitatively different conclusion (majority-integrated, $\tau_{\text{int}} \sim \tau_c$) from the same readings. The linear link is the empirically favoured one, so $w \approx 0.23$ is the current best guess, but the interpretation of w is not robust to the link, and the link is settled on bulk data.

7. Fitting

Data. One row per molecule: n (v5 `n_tf_teto`), the promoter activity call (v5 region-anchored `{region}_open / {region}_nuc`), and the molecule’s condition key (K , dox, AD, background). Join $\langle n \rangle$ on by condition.

Objective. With the identity link the model is a **linear probability model**,

$$P = p_0 + k_{\text{fast}} n + k_{\text{slow}} \langle n \rangle,$$

so it fits by ordinary (or molecule-count-weighted) least squares — a closed form, no optimizer, no convergence question, and the same WLS family `compute_potency.py` already uses. This is a side benefit of §2.1’s empirical finding: the link that fits best is also the one that makes the estimator trivial.

Two standard caveats of linear probability models apply and are both benign here: predictions are not guaranteed to lie in $[0, 1]$ (harmless while $\bar{P} \lesssim 0.5$, but check, and revisit if a strong new AD pushes activity high), and the residuals are heteroscedastic, so use robust or $\sqrt{P(1-P)}$ -weighted standard errors rather than the naive OLS ones. If either becomes a problem, switch to the hyperbolic link and a bounded nonlinear fit — note that link’s likelihood is *not* concave ($\partial^2/\partial A^2$ of $y \log A - \log(1 + A)$ is $-y/A^2 + 1/(1 + A)^2$, negative for $y = 1$ but positive for $y = 0$), so it needs multi-start.

Identifiability. k_{fast} comes from *within*-condition variation in n ; k_{slow} from *between*-condition variation in $\langle n \rangle$. Both are needed. A single condition identifies only k_{fast} ; bulk data alone identifies only k_{pot} . The copy-number series and the dox series each supply the between-condition axis, and having both allows a consistency check: **w estimated from the copy-number series and from the dox series should agree.** If they do not, the “history” variable is not simply time-averaged occupancy.

Hierarchy. Same discipline as elsewhere: p_0 per promoter (sequence differences are real); k_{pot} per AD (that is the scientific quantity); w shared across ADs to start — it is a timescale ratio, so per-AD variation is a claim needing evidence. Freeing w per AD is the interesting extension (an AD

acting through a stably recruited cofactor should have larger w than one acting by transient PIC recruitment), but it should be earned by model comparison, not assumed.

Do not use --background None. The runs of record pooled opJS4 (b0) and opJS5 (b1/b2) into one regression per sample. Fig. 3b shows the backgrounds differ substantially in activity at matched copy number, so p_0 at least must be indexed by background here.

8. Mechanism: what is doing the integrating?

If $w > 0$, some slow variable carries memory of past binding. The leading candidate is **chromatin**, and it is testable with existing data.

The old `3param_nuc` model already encodes the relevant coupling: it shifts the nucleosome energy whenever any TF is bound,

```
num_tfs * tf_e + num_nuc * (nuc_e - delta_e * (num_tfs > 0))
```

i.e. a recruited-remodeler term. If a remodeler evicts a nucleosome and the depleted state persists longer than TF residence, then the nucleosome *is* the integrator, and τ_{int} is a nucleosome re-formation time.

Test: run the §5.1/§5.2 analysis on **nucleosome occupancy** rather than promoter activity.

- If $P(\text{nuc} | n)$ shows the same $\langle n \rangle$ -dependence, the memory is in the chromatin layer — and δ_{remodel} , fitted in the equilibrium array model, becomes the mechanistic parameter behind w .
- If nucleosome occupancy shows *no* $\langle n \rangle$ -dependence while promoter activity does, the integrator is downstream of chromatin (cofactor residence, PIC assembly, a modification).

Either outcome identifies the physical carrier of the memory, which is the mechanistic basis for linear potency that the original analysis could not supply.

9. What this model costs

Stated plainly, because it is a real conceptual price:

- **It is not a thermodynamic model.** $\langle n \rangle$ is a property of the *ensemble*, not of the configuration, so no Boltzmann distribution over single-molecule states can produce this conditional. k_{pot} and w are not free energies and should not be reported in $k_B T$. The reciprocity argument (TF stabilizes the promoter and the promoter stabilizes TF binding via one symmetric coupling) applies only to the $w = 0$ limit, which the data reject.
- **The equilibrium array model is unaffected.** TetO occupancy, cooperativity J , nucleosome fugacity μ , and δ_{remodel} all remain a legitimate equilibrium lattice-gas problem (260717_ising_intuition.md §§1–6, §8). Only the *promoter* node leaves equilibrium. The clean architecture is therefore two stages: an equilibrium array model that predicts $\langle n \rangle$ and the per-molecule distribution of n , and a kinetic promoter model on top of it. §7 of the intuition doc — the promoter as one more Ising node — should be treated as superseded by this note.

- **Snapshots cannot see timescales directly.** τ_{int}/τ_c is inferred from the *ratio* $w/(1-w)$ under the linear-filter assumption of §3.1, not measured. Converting it to seconds requires an independent estimate of rTetR residence time.
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10. Checklist

1. **Confirm the link per AD, not pooled** (§2.1). $\bar{P}$ vs $\langle n \rangle$ should be straight; the interesting case is a strong new AD that reaches high $\bar{P}$ and either stays straight (a real result) or bends (the bend identifies the link). Pooling ADs averages this away.
2. **Re-plot 3h as the model wants it** (§5.1): P vs $\langle n \rangle_K$ at fixed n . Check the lines are parallel. Read off $k_{\text{pot}}w$ (common slope) and $k_{\text{pot}}(1-w)$ (spacing), and check that k_{pot} agrees with the Fig. 3d slope (§6).
3. **Run the dox test** (§5.2): $P(\text{active} | n)$ vs dox at fixed K . This is the confound-free test for $w > 0$.
4. **Cross-check** that w from the copy-number and dox series agree (§7).
5. **Condition on exact configuration** (§5.3) using v5 per-site calls.
6. **Ask whether the memory is chromatin** (§8) by repeating 2–3 on nucleosome occupancy.
7. Only then fit the full model, hierarchically, with backgrounds separated (§7).

Steps 1–3 need no new machinery — they are re-tabulations of existing molecule tables — and their outcomes determine whether steps 4–7 are worth building.